



Babelfish

CPU TIME LIMIT

3 seconds

MEMORY LIMIT

1024 MB

DIFFICULTY

easy (2)

You have just moved from Waterloo to a big city. The people here speak an incomprehensible dialect of a foreign language. Fortunately, you have a dictionary to help you understand them.

Input

Input consists of up to 100 000 dictionary entries, followed by a blank line, followed by a message of up to 100 000 words. Each dictionary entry is a line containing an English word, followed by a space and a foreign language word. No foreign word appears more than once in the dictionary. The message is a sequence of words in the foreign language, one word on each line. Each word in the input is a non-empty sequence of at most 10 lowercase letters.

Output

Output is the message translated to English, one word per line. Foreign words not in the dictionary should be translated as “eh”.



Sample Input 1

```
dog ogday  
cat atcay  
pig igpay  
froot ootfray  
loops oopslay  
  
atcay  
ittenkay  
oopslay
```

Sample Output 1

```
cat  
eh  
loops
```